

# Chapter 3

## Complex Integration

**Definition 3.1.** The *primitive* of a function  $f : \mathcal{D} \rightarrow \mathbb{C}$  is an analytic function  $F : \mathcal{D} \rightarrow \mathbb{C}$  such that  $F' = f$  for any  $z \in \mathcal{D}$ .

**Motivation 3.2.** In the case of  $\mathcal{D} \subseteq \mathbb{R}$ , the first fundamental theorem of calculus (Theorem B.127) guarantees the existence of the primitive of any function  $f : \mathcal{D} \rightarrow \mathbb{R}$ , since we can construct a suitable integral as the primitive of  $f$ . However, this is not true in the case of  $\mathcal{D}$  being an open subset of  $\mathbb{C}$ : a function that has a primitive must be analytic, c.f. Theorem 3.83. Therefore, it is natural to ask the question whether the primitive of a function  $f : \mathcal{D} \rightarrow \mathbb{C}$  exists; this question is a key problem in our extension of  $\mathbb{R}$  to  $\mathbb{C}$ .

**Roadmap 3.3.** This chapter mainly concerns the equivalence of the following four statements.

- (a)  $f$  has a primitive.
- (b) The complex integral of  $f$  on a closed curve vanishes.
- (c) The complex line integral of  $f$  is path independent.
- (d)  $f$  is analytic.

After examining a stepping-stone integral in Definition 3.4, we formally define the concept of complex line integrals in Section 3.2 and prove Theorem 3.16 that states (a)  $\Rightarrow$  (b). One key result of Section 3.3 is Theorem 3.40, which states that (a)  $\Leftrightarrow$  (b)  $\Leftrightarrow$  (c). The central result of Section 3.3 is the Cauchy theorem, which states (d)  $\Rightarrow$  (b) in Theorems 3.45, 3.53, 3.57, 3.62, and 3.71 on various domains. In Section 3.4, the generalized Cauchy integral formula (Theorem 3.83) gives (a)  $\Rightarrow$  (d); this is utilized in the proof of the Morera theorem 3.91, which states (b)  $\Rightarrow$  (d).

### 3.1 A stepping stone

**Definition 3.4.** A function  $f : [a, b] \rightarrow \mathbb{C}$  with  $a, b \in \mathbb{R}$  is called *integrable* iff  $\operatorname{Re} f, \operatorname{Im} f : [a, b] \rightarrow \mathbb{R}$  are Riemann integrable as in Definition B.106; then

$$\int_a^b f(x) dx = \int_a^b \operatorname{Re} f(x) dx + i \int_a^b \operatorname{Im} f(x) dx. \quad (3.1)$$

**Corollary 3.5.** Denote by  $\mathcal{V}$  the set of continuous functions  $[a, b] \rightarrow \mathbb{C}$ . Then  $\mathcal{V}$  is a vector space and the integral  $\int_a^b$  defined in (3.1) is a linear functional on  $\mathcal{V}$ .

*Proof.* We need to show that the integration is a linear map from  $\mathcal{V}$  to  $\mathbb{C}$ , which holds because

$$\begin{aligned} \int_a^b (f(x) + g(x)) dx &= \int_a^b f(x) dx + \int_a^b g(x) dx; \\ \forall \lambda \in \mathbb{C}, \int_a^b \lambda f(x) dx &= \lambda \int_a^b f(x) dx. \end{aligned} \quad \square$$

**Lemma 3.6.** If  $f : [a, b] \rightarrow \mathbb{C}$  is continuous and  $F$  is a primitive of  $f$ , i.e.  $F' = f$ , then

$$\int_a^b f(x) dx = F(b) - F(a). \quad (3.2)$$

*Proof.* This follows from (3.1) and the second fundamental theorem of calculus (Theorem B.128).  $\square$

**Lemma 3.7.** If  $f : [a, b] \rightarrow \mathbb{C}$  is integrable and satisfies  $|f(x)| \leq C$  for all  $x \in [a, b]$ , then we have

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx \leq (b-a)C. \quad (3.3)$$

*Proof.* The second inequality follows from the integral mean value theorem B.124. To prove the first inequality, define  $\int_a^b f(x) dx = re^{i\theta}$  and we have

$$\begin{aligned} \left| \int_a^b f(x) dx \right| &= r = e^{-i\theta} \int_a^b f(x) dx = \int_a^b e^{-i\theta} f(x) dx \\ &= \operatorname{Re} \int_a^b e^{-i\theta} f(x) dx = \int_a^b \operatorname{Re} [e^{-i\theta} f(x)] dx \\ &\leq \int_a^b |e^{-i\theta} f(x)| dx = \int_a^b |f(x)| dx, \end{aligned}$$

where all steps follows from complex arithmetic.  $\square$

**Example 3.8** (Lemma 3.7 as a special case of the Minkowski inequality in  $L^p$  space). The first inequality in (3.3) is a direct consequence of the Minkowski inequality (Theorem F.11) on the  $L^p$  space,

$$\forall p \geq 1, \quad \|f + g\|_p \leq \|f\|_p + \|g\|_p, \quad (3.4)$$

where  $f, g \in L^p(\Omega)$  and

$$\|f\|_p = \left( \int_{\Omega} |f(x)|^p dx \right)^{\frac{1}{p}}. \quad (3.5)$$

This inequality is important in establishing that  $(L^p, \|\cdot\|_p)$  is a normed linear space for  $p \geq 1$ . For  $p < 1$ , (3.4) holds for non-negative functions  $f, g$  with the inequality reversed:

$$\forall p < 1, \quad \|f + g\|_p \geq \|f\|_p + \|g\|_p. \quad (3.6)$$

Then Lemma 3.7 follows from setting  $f = u^2$ ,  $g = v^2$ ,  $p = \frac{1}{2}$  in (3.6) and using the inequality  $\int_a^b u \leq \int_a^b |u|$ .

**Theorem 3.9.** If  $f : [a, b] \rightarrow \mathbb{R}^k$  is integrable, then  $\|f\|$  is integrable and

$$\left\| \int_a^b f dx \right\| \leq \int_a^b \|f\| dx, \quad (3.7)$$

where  $\|\cdot\|$  denotes the Euclidean norm in Definition C.176.

*Proof.* Write  $f(x) := (f_1(x), \dots, f_k(x))$ ,  $y_i := \int_a^b f_i(x) dx$ , and we have  $y := \int_a^b f(x) dx = (y_1, \dots, y_k)$  and

$$\|y\|^2 = \sum y_j^2 = \sum y_j \int_a^b f_j dx. \quad (3.8)$$

For a fixed  $x \in [a, b]$ , the Cauchy-Schwarz inequality yields

$$\langle y, f(x) \rangle = \sum y_i f_i(x) \leq \|y\| \|f(x)\|.$$

Integrate the above inequality, apply (3.8), and we have

$$\|y\|^2 = \sum y_j \int_a^b f_j dx \leq \|y\| \int_a^b \|f(x)\| dx.$$

If  $\|y\| \neq 0$ , we divide both sides of the above inequality by  $\|y\|$  to obtain (3.7). If  $\|y\| = 0$ , each  $y_i$  must be zero, hence (3.7) also holds trivially.  $\square$

**Lemma 3.10.** For continuously differentiable functions  $u, v : [a, b] \rightarrow \mathbb{C}$ , we have

$$\int_a^b u(x)v'(x)dx = (uv)\Big|_a^b - \int_a^b u'(x)v(x)dx. \quad (3.9)$$

*Proof.* This follows from Lemma 3.6 and the product rule  $(uv)' = uv' + u'v$  in Theorem 2.20.  $\square$

**Lemma 3.11.** Let  $I_1, I_2 \subset \mathbb{R}$  be intervals,  $a, b \in I_1$  and  $\varphi : I_1 \rightarrow I_2$  continuously differentiable and  $f : I_2 \rightarrow \mathbb{C}$  continuous. Then

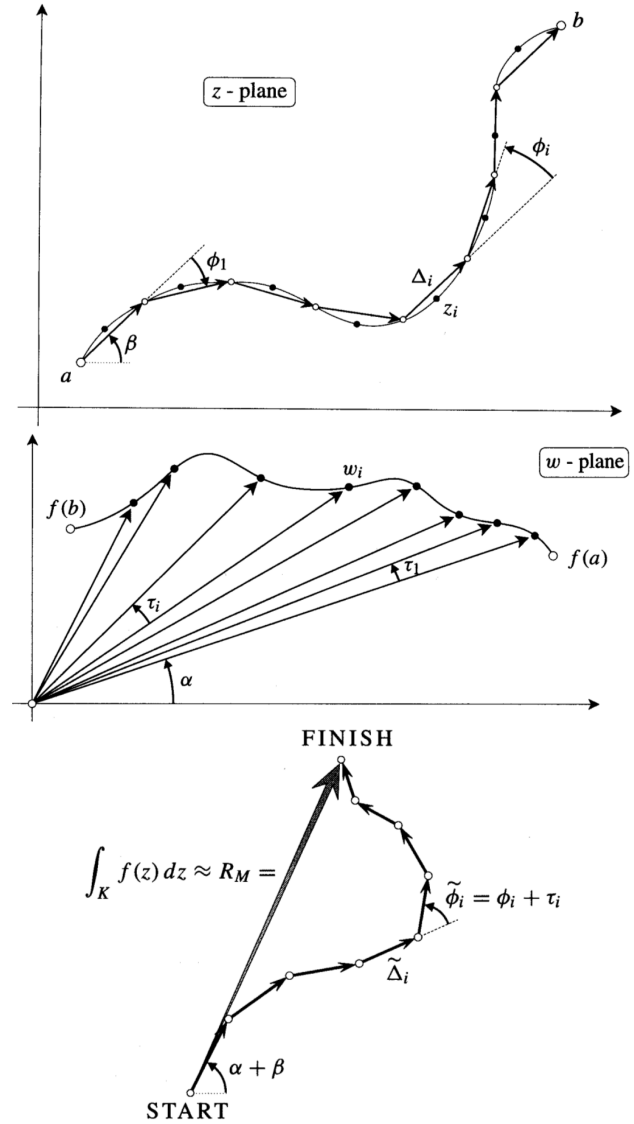
$$\int_{\varphi(a)}^{\varphi(b)} f(y)dy = \int_a^b f(\varphi(x))\varphi'(x)dx. \quad (3.10)$$

*Proof.* This follows directly from Theorem B.130 and Definition 3.4.  $\square$

## 3.2 Complex line integrals

**Definition 3.12.** For a piecewise smooth curve  $K : [a, b] \rightarrow \mathbb{C}$  and a continuous function  $f : \mathcal{D} \rightarrow \mathbb{C}$  with  $\mathcal{D} \supset K([a, b])$ , the *complex line integral* or *complex contour integral* of  $f$  along  $K$ , written  $\int_K f$ , is given by

$$\int_K f(z)dz := \int_a^b f(K(t))K'(t)dt. \quad (3.11)$$



**Lemma 3.13.** Let  $\alpha : [c, d] \rightarrow \mathbb{C}$  be a piecewise smooth curve and  $f : \mathcal{D} \rightarrow \mathbb{C}$  be a continuous function with  $\alpha([c, d]) \subset \mathcal{D} \subset \mathbb{C}$ . Let  $\varphi : [a, b] \rightarrow [c, d]$  be a continuously differentiable function with  $\varphi(a) = c$  and  $\varphi(b) = d$ . Then we have

$$\int_{\alpha} f(\zeta)d\zeta = \int_{\alpha \circ \varphi} f(\zeta)d\zeta. \quad (3.12)$$

*Proof.* Definition 3.12 yields

$$\begin{aligned} \int_{\alpha} f(\zeta)d\zeta &= \int_c^d f(\alpha(t))\alpha'(t)dt \\ &= \int_a^b f(\alpha(\varphi(s)))\alpha'(\varphi(s))d\varphi(s) \\ &= \int_a^b f[(\alpha \circ \varphi)(s)](\alpha \circ \varphi)'(s)ds = \int_{\alpha \circ \varphi} f(\zeta)d\zeta. \quad \square \end{aligned}$$

**Lemma 3.14.** Let  $f : \mathcal{D} \rightarrow \mathbb{C}$  be a continuous function with  $\mathcal{D} \subset \mathbb{C}$  open. If  $f$  has a primitive  $F$ , i.e.  $F' = f$ , then for any piecewise smooth curve  $\alpha \subset \mathcal{D}$  we have

$$\int_{\alpha} f(\zeta) d\zeta = F(\alpha(b)) - F(\alpha(a)). \quad (3.13)$$

*Proof.* Let  $F = U + iV$ , then the Cauchy–Riemann equations yield  $f = \frac{\partial U}{\partial x} + i \frac{\partial V}{\partial x} = \frac{\partial V}{\partial y} - i \frac{\partial U}{\partial y}$ . Write  $\alpha(t) = x(t) + iy(t)$  and we have

$$\begin{aligned} f(\alpha(t))\alpha'(t) &= \frac{\partial U}{\partial x}x' + \frac{\partial U}{\partial y}y' + i \left( \frac{\partial V}{\partial x}x' + \frac{\partial V}{\partial y}y' \right) \\ &= \frac{dU}{dt} + i \frac{dV}{dt}. \end{aligned}$$

Then (3.13) follows from integrating the above identity over  $\alpha$  and applying (3.11) and the fundamental theorem of calculus (Theorem B.128).  $\square$

**Corollary 3.15.**  $\forall z \in \mathbb{C}_-, \text{Log}(z) = L(z) := \int_1^z \frac{1}{\zeta} d\zeta$ .

*Proof.* Corollary 2.26 states that  $\text{Log}(z)$  is analytic in  $\mathbb{C}_-$  and  $(\text{Log}(z))' = \frac{1}{z}$ . The rest follows from Lemma 3.14.  $\square$

**Theorem 3.16.** Let  $f : \mathcal{D} \rightarrow \mathbb{C}$  be a continuous function with  $\mathcal{D} \subset \mathbb{C}$  open. If  $f$  has a primitive  $F$ , i.e.  $F' = f$ , then for any closed piecewise smooth curve  $\gamma$  in  $\mathcal{D}$  we have

$$\oint_{\gamma} f(\zeta) d\zeta = 0. \quad (3.14)$$

*Proof.* Decompose  $\gamma$  into two curves  $\alpha, \beta : [0, 1] \rightarrow \mathbb{C}$  such that  $\gamma = \alpha \cup \beta$ ,  $\alpha(0) = \beta(1)$ , and  $\alpha(1) = \beta(0)$ . The proof is completed by  $\oint_{\gamma} = \int_{\alpha} + \int_{\beta}$  and Lemma 3.14.  $\square$

**Corollary 3.17.** For a counterclockwise oriented circle  $\gamma(t) = r \exp(it)$  where  $r > 0$  and  $t \in [0, 2\pi]$ , we have

$$\forall n \in \mathbb{Z}, \quad \oint_{\gamma} \zeta^n d\zeta = \begin{cases} 2\pi i & \text{if } n = -1, \\ 0 & \text{otherwise.} \end{cases} \quad (3.15)$$

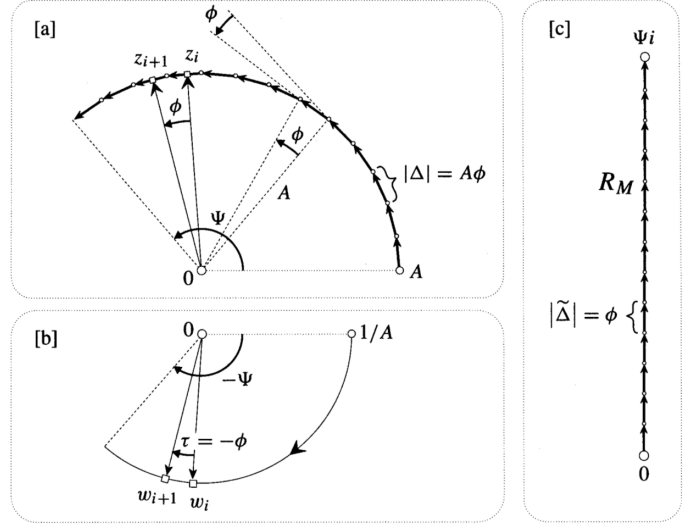
*Proof.* For  $n \neq -1$ , the function  $f(z) = z^n$  has the primitive  $F(z) = \frac{z^{n+1}}{n+1}$ , c.f. Exercise 2.22. For  $n = -1$ ,

$$\oint_{\gamma} \zeta^{-1} d\zeta = \int_0^{2\pi} (re^{it})^{-1} r i e^{it} dt = 2\pi i. \quad \square$$

**Exercise 3.18** (An alternative proof of Corollary 3.17 via Corollary 3.15). By Corollary 2.26, we have  $(\text{Log}z)' = \frac{1}{z}$  for  $z \in \mathbb{C}_-$ . By Definition 3.24, the unit circle  $\mathbb{S}^1 = \alpha \oplus \beta$  is the concatenation of two curves  $\theta \mapsto e^{i\theta}$ ,

$$\alpha := ((-\pi)^+, 0) \rightarrow \mathbb{C}, \quad \beta := (0, \pi) \rightarrow \mathbb{C}.$$

Put  $z_{-1}^- := \alpha((-\pi)^+)$  and  $z_{-1}^+ := \beta(\pi)$ . Based on the above constructions, starting from the variable substitution  $\zeta = rz$ , calculate  $\oint_{\gamma} \zeta^{-1} d\zeta$ .



**Corollary 3.19.** The function  $f(z) = \frac{1}{z}$  does not have a primitive if its domain is  $\mathbb{C}^*$ .

*Proof.* Otherwise it would contradict Corollary 3.17.  $\square$

**Exercise 3.20.** Draw plots similar to those of  $\int_K \frac{1}{z} dz$  to give a geometric interpretation of  $\int_A^B z^m dz$  where  $m \in \mathbb{N}^+$  and the integral is on the arc  $\gamma(t) = r \exp(it)$  with  $t \in [\text{Arg}A, \text{Arg}B]$ . Then deduce the formula

$$\int_A^B z^m dz = \frac{1}{m+1} (B^{m+1} - A^{m+1}). \quad (3.16)$$

**Exercise 3.21.** Give a geometric interpretation of  $\int_A^B e^z dz$  and the integral is on the vertical segment  $\gamma(t) = A + it$  with  $t \in [0, \theta]$ , where  $B = A + i\theta$  and  $\theta > 0$ . Then deduce the formula

$$\int_A^B e^z dz = e^B - e^A. \quad (3.17)$$

**Exercise 3.22.** In Corollary 3.17, we computed the integral  $\oint_{\gamma} \zeta^n d\zeta$  for all integers  $n$ , where  $\gamma$  is a circle centered at the origin with positive orientation. Without invoking the Cauchy integral theorem, calculate  $\oint_{\gamma} \zeta^n d\zeta$  where

- (a)  $\gamma$  is any circle not enclosing the origin;
- (b)  $\gamma$  is a positively oriented square with vertices at  $\pm 1 \pm i$ .

*Hint:* For each of the four line segments, use the parameterization and calculate the real integral  $\int_{-1}^1 \frac{1}{1+t^2} dt$ .

**Exercise 3.23.** For the counterclockwise oriented circle  $\gamma(t) = r \exp(it)$  with  $t \in [0, 2\pi]$  and  $r \in (|a|, |b|)$ , prove

$$\oint_{\gamma} \frac{1}{(\zeta - a)(\zeta - b)} d\zeta = \frac{2\pi i}{a - b}.$$

*Hint:* Express  $\frac{1}{\zeta - a}$  as the geometric series expansion  $\frac{1}{1 - z} = \sum_{n=0}^{\infty} z^n$  for  $|z| < 1$ .

### 3.3 The Cauchy integral theorem

#### 3.3.1 Simply connected domains

**Definition 3.24.** The *concatenation of two curves*  $\alpha : [a, b] \rightarrow \mathbb{C}$  and  $\beta : [b, c] \rightarrow \mathbb{C}$  with  $\alpha(b) = \beta(b)$  is another curve  $\alpha \oplus \beta : [a, c] \rightarrow \mathbb{C}$  given by

$$(\alpha \oplus \beta)(t) := \begin{cases} \alpha(t) & \text{if } t \in [a, b], \\ \beta(t) & \text{if } t \in [b, c]. \end{cases} \quad (3.18)$$

**Example 3.25.**  $\alpha(t) = \sqrt{1-t^2}$ ,  $t \in [-1, 0]$  and  $\beta(t) = 1$ ,  $t \in [0, 2]$  are two smooth curves. The concatenation  $\alpha \oplus \beta$  is not smooth but piecewise smooth: the second derivatives of  $\alpha$  and  $\beta$  are different at  $t = 0$ . Indeed, the curvature of a unit circle and a straight line is 1 and 0, respectively.

**Definition 3.26.** The *reciprocal curve* of a curve  $\alpha : [a, b] \rightarrow \mathbb{C}$  is the curve  $\alpha^-(t) : [a, b] \rightarrow \mathbb{C}$  defined by  $\alpha^-(t) = \alpha(b + a - t)$ .

**Definition 3.27.** A set  $\mathcal{D} \subset \mathbb{C}$  is called *arcwise connected* or *path connected* iff for any two points  $z, w \in \mathcal{D}$  there is a piecewise smooth curve  $\alpha : [a, b] \rightarrow \mathcal{D}$  such that  $\alpha(a) = z$  and  $\alpha(b) = w$ . In particular, an arcwise connected nonempty open set is called a *domain*.

**Example 3.28.** The point set  $\mathcal{D} = \{z \in \mathbb{C} : |z^2 - 3| < 1\}$  is not a domain because it is not connected:  $\pm\sqrt{3} \in \mathcal{D}$  whereas  $\{yi : y \in \mathbb{R}\} \cap \mathcal{D} = \emptyset$ .

**Example 3.29.** The set  $\mathcal{D} \setminus \mathcal{D}'$  is a domain where

$$\mathcal{D} = \{x + iy : x \in (0, 1), y \in (0, 1)\};$$

$$\mathcal{D}' = \cup_{n=2}^{\infty} \left\{ x + iy : x = \frac{1}{n}, y \in \left(0, \frac{1}{2}\right] \right\},$$

because  $\mathcal{D} \setminus \mathcal{D}' = \mathcal{D} \setminus \overline{\mathcal{D}'} = \mathcal{D} \cap (\mathbb{C} \setminus \overline{\mathcal{D}'}): \overline{\mathcal{D}'}$  is closed,  $(\mathbb{C} \setminus \overline{\mathcal{D}'})$  is open, and thus  $\mathcal{D} \setminus \mathcal{D}'$  is open.  $\mathcal{D} \setminus \mathcal{D}'$  is also path connected.

**Definition 3.30.** A *path* in a topological space  $\mathcal{X}$  is a continuous map  $\zeta : [0, 1] \rightarrow \mathcal{X}$ .

**Definition 3.31.** A *loop* or *cycle* is a path whose endpoints coincide; this common endpoint is the *basepoint* of the loop.

**Definition 3.32.** A *homotopy of paths* in  $\mathcal{X}$  is a family of paths  $h_t : [0, 1] \rightarrow \mathcal{X}$ ,  $t \in [0, 1]$ , such that  $h_t(0)$ ,  $h_t(1)$  are independent of time and the associated map  $H : [0, 1]^2 \rightarrow \mathcal{X}$  defined by  $H(s, t) = h_t(s)$  is continuous. Two paths are *homotopic* if they are connected by a homotopy.

**Example 3.33.** The family of flow maps of a well-posed ODE  $\frac{dx}{dt} = \mathbf{u}(\mathbf{x}, t)$  that act on a path at  $t = 0$  is a homotopy of paths in the phase space of the ODE.

**Lemma 3.34.** The relation of homotopy on paths with fixed endpoints in any space is an equivalence relation.

*Proof.* This follows from Definitions 3.32 and A.36.  $\square$

**Definition 3.35.** A *free homotopy* between two loops  $\gamma_1$  and  $\gamma_2$  in  $\mathcal{X}$  is a function  $H_f : [0, 1]^2 \rightarrow \mathcal{X}$  such that  $H_f(s, 0) = \gamma_1(s)$  and  $H_f(s, 1) = \gamma_2(s)$  for all  $s$ , and  $H_f(0, t) = H_f(1, t)$  for all  $t$ . Then  $\gamma_1$  and  $\gamma_2$  are said to be *freely homotopic* in  $\mathcal{X}$ .

**Example 3.36.** If the basepoint of a free loop homotopy is further required to be fixed, i.e.,  $H_f(0, t) = H_f(1, t) = x_0$  for all  $t \in [0, 1]$ , we get a *based homotopy* or a *path homotopy*, in which case we also say that the two loops are freely homotopic relative to their endpoints.

**Example 3.37.** In the space of  $\mathbb{S}^1 = [0, 1]/\{0, 1\}$ , the loops  $\alpha(s) := e^{2\pi i s}$  and  $\beta(s) := 1$  are freely homotopic through  $H_f(s, t) = e^{2\pi i s(1-t)}$ . If the basepoint at  $t = 0$  is chosen to be  $x_*(0) = \alpha(1)$ , then the expression of the basepoint at time  $t$  is  $x_*(t) = e^{2\pi i(1-t)}$ . During this homotopy,  $x_*$  covers  $\mathbb{S}^1$  once. However, if  $x_*$  is required to be fixed at  $\alpha(1)$ , the loop  $\alpha$  cannot be deformed into the constant loop  $\beta$ .

**Definition 3.38.** A topological space  $\mathcal{X}$  is *simply connected* if and only if it is path connected and whenever  $\alpha : [0, 1] \rightarrow \mathcal{X}$  and  $\beta : [0, 1] \rightarrow \mathcal{X}$  are two paths with the same start and end points ( $\alpha(0) = \beta(0)$  and  $\alpha(1) = \beta(1)$ ), then there exists a based homotopy  $H : [0, 1]^2 \rightarrow \mathcal{X}$  such that  $H(s, 0) = \alpha(s)$  and  $H(s, 1) = \beta(s)$ , i.e.,  $\alpha$  can be continuously deformed into  $\beta$  while keeping both endpoints fixed.

**Exercise 3.39.** Prove that Theorem 2.73 holds not only for rectangular domains but also for simply connected domains.

#### 3.3.2 A proof by Riemann

**Theorem 3.40.** For a continuous function  $f : \mathcal{D} \rightarrow \mathbb{C}$  with  $\mathcal{D}$  a domain, the following statements are equivalent:

- (a)  $f$  has a primitive.
- (b) The integral of  $f$  along any piecewise smooth closed curve in  $\mathcal{D}$  vanishes.
- (c) The integral of  $f$  along any piecewise smooth curve in  $\mathcal{D}$  depends only on the beginning and end points of the curve.

*Proof.* By Theorem 3.16, (a) implies (b).

To show (b) implies (c), let  $\alpha : [a, b] \rightarrow \mathcal{D}$  and  $\beta : [c, d] \rightarrow \mathcal{D}$  be two curves with the same starting and ending points. WLOG we can assume  $b = c$ . Otherwise set  $\varphi(t) = t + c - b$  in Lemma 3.13 and we have  $\int_{\beta} f(\zeta) d\zeta = \int_{\beta \circ \varphi} f(\zeta) d\zeta$ . It follows that

$$0 = \oint_{\alpha \oplus \beta^-} f = \int_{\alpha} f - \int_{\beta} f.$$

To show (c) implies (a), fix a point  $z_* \in \mathcal{D}$  and construct the integral

$$F(z) = \int_{z_*}^z f(\zeta) d\zeta, \quad (3.19)$$

where the integral is along a piecewise smooth curve connecting  $z_*$  with  $z$  in  $\mathcal{D}$ . It follows from Lemma 2.18 that  $F'(z_0) = f(z_0)$  holds for any given  $z_0 \in \mathcal{D}$  if the residual

$$r(z) := F(z) - F(z_0) - f(z_0)(z - z_0)$$

satisfies

$$\lim_{z \rightarrow z_0} \frac{r(z)}{|z - z_0|} = 0. \quad (3.20)$$

Since  $z_0 \in \mathcal{D}$ , there exists an open ball  $\mathcal{U}_\rho(z_0)$  that is a subset of  $\mathcal{D}$ . For all  $z \in \mathcal{U}_\rho(z_0)$ , we have

$$F(z) = \int_{z_*}^{z_0} f(\zeta) d\zeta + \int_{z_0}^z f(\zeta) d\zeta = F(z_0) + \int_{z_0}^z f(\zeta) d\zeta,$$

where the last integral can be taken from  $z_0$  to  $z$  along the line segment  $\sigma : [0, 1] \rightarrow \mathcal{D}$  given by  $\sigma(t) := z_0 + t(z - z_0)$ . Definition 3.12 yields  $\int_\sigma d\zeta = z - z_0$ . Hence we have

$$r(z) = \int_\sigma (f(\zeta) - f(z_0)) d\zeta$$

and the continuity of  $f$  gives

$$\forall \epsilon > 0, \exists \delta \in (0, \rho), \text{ s.t. } |z - z_0| < \delta \Rightarrow |f(z) - f(z_0)| < \epsilon.$$

Then Lemma 3.7 yields (3.20). The proof is completed by  $z_0$  being arbitrary.  $\square$

**Exercise 3.41.** Determine whether the following functions have primitives on the indicated domains and fully justify your answer.

- (a)  $f(z) = \frac{1}{z}$  on  $\mathbb{C}^\bullet$ ;
- (b)  $g(z) = \bar{z}$  on  $\mathbb{C}$ .

**Exercise 3.42.** Let  $\gamma : [0, 1] \rightarrow \mathbb{H} := \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$  be a piecewise smooth curve with  $\gamma(0) = 1 + \mathbf{i}$  and  $\gamma(1) = \mathbf{i}$ .

- (a) Do  $z$  and  $\frac{1}{z}$  have primitives on  $\mathbb{H}$ ? Justify your answer.
- (b) Calculate  $\int_\gamma z dz$  and  $\int_\gamma \frac{1}{z} dz$ .

**Theorem 3.43** (Green). Let  $\alpha$  be a counter-clockwise oriented, piecewise smooth, simple closed curve in the plane. Denote by  $\mathcal{D}$  the bounded complement of  $\alpha$  and  $u, v : \mathcal{D} \rightarrow \mathbb{R}$  functions with continuous partial derivatives on  $\mathcal{D}$ . Then

$$\oint u(x, y) dx + v(x, y) dy = \iint \left( \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx dy. \quad (3.21)$$

**Lemma 3.44.** If  $f(z) = u(x, y) + \mathbf{i}v(x, y)$  is continuous along a piecewise smooth curve  $\alpha$ , then  $f(z)$  is integrable along  $\alpha$  and

$$\int_\alpha f(z) dz = \int_\alpha u dx - v dy + \mathbf{i} \int_\alpha v dx + u dy. \quad (3.22)$$

*Proof.* This follows from  $dz = dx + \mathbf{i}dy$ , Definition 3.4, and Theorem B.119 that a continuous function is integrable.  $\square$

**Theorem 3.45** (Cauchy). If  $f : \mathcal{D} \rightarrow \mathbb{C}$  is analytic in a simply connected domain  $\mathcal{D}$  and  $f'$  is continuous in  $\mathcal{D}$ , then

$$\oint_{\partial \mathcal{D}} f(\zeta) d\zeta = 0. \quad (3.23)$$

*Proof.* Let  $f(z) = u(x, y) + \mathbf{i}v(x, y)$  and  $z = x + \mathbf{i}y$ . In 1851, Riemann reasoned that

$$\begin{aligned} \oint_{\partial \mathcal{D}} f(\zeta) d\zeta &= \oint_{\partial \mathcal{D}} u dx - v dy + \mathbf{i} \oint_{\partial \mathcal{D}} v dx + u dy \\ &= \iint_{\mathcal{D}} -\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right) dx dy + \mathbf{i} \iint_{\mathcal{D}} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dx dy = 0, \end{aligned}$$

where the first step follows from Lemma 3.44, the second from Theorem 3.43, and the third from Theorem 2.50.  $\square$

**Lemma 3.46** (Polynomial growth). For a complex polynomial of degree  $n \in \mathbb{N}^+$ ,

$$P(z) = \sum_{j=0}^n a_j z^j, \quad (3.24)$$

where  $a_j \in \mathbb{C}$  for all  $j = 0, \dots, n$  and  $a_n \neq 0$ , we have

$$\frac{1}{2}|a_n||z|^n \leq |P(z)| \leq \frac{3}{2}|a_n||z|^n \quad (3.25)$$

for any  $z$  with a sufficiently large modulus  $|z| \geq \rho$ , e.g.,

$$\rho := \max \left\{ 1, \frac{2}{|a_n|} \sum_{j=0}^{n-1} |a_j| \right\}. \quad (3.26)$$

*Proof.* Define  $Q(z) = P(z) - a_n z^n$ . Then the triangle inequality and the condition  $|z| \geq \rho$  yield

$$|Q(z)| \leq \left( \sum_{j=0}^{n-1} |a_j| \right) |z|^{n-1} \leq \frac{|a_n|}{2} |z|^n.$$

Apply the triangle inequality, i.e. Corollary 1.66, again to  $Q(z) + a_n z^n = P(z)$  and we have

$$|a_n||z|^n - |Q(z)| \leq |P(z)| \leq |a_n||z|^n + |Q(z)|.$$

Combining the two inequalities yields (3.25).  $\square$

**Theorem 3.47** (Fundamental theorem of Algebra). Any nonconstant complex polynomial (3.24) has a root.

*Proof.* Suppose  $P(z) \neq 0$  for all  $z \in \mathbb{C}$ . Define  $Q(z)$  by  $P(z) = zQ(z) + a_0$ . Then for any  $z \neq 0$  we have

$$\frac{1}{z} = \frac{zQ(z) + a_0}{zP(z)} = \frac{Q(z)}{P(z)} + \frac{a_0}{zP(z)}.$$

By Corollary 3.17 and Theorem 3.45, the integral of  $\frac{1}{z}$  along the circle  $\alpha(t) = R \exp(\mathbf{i}t)$  with  $t \in [0, 2\pi]$  and  $R > 0$  yields

$$2\pi \mathbf{i} = \oint_\alpha \frac{a_0}{zP(z)} dz,$$

where we have used the facts of  $P(z) \neq 0$  and  $P/Q$  being analytic. Then Lemma 3.7 and Lemma 3.46 yield

$$2\pi = \left| \oint_\alpha \frac{a_0}{zP(z)} dz \right| \leq \oint_\alpha \left| \frac{a_0}{zP(z)} \right| dz \leq 2\pi R \frac{2|a_0|}{R|a_n|R^n},$$

which cannot hold for any sufficiently large  $R > 0$ .  $\square$

### 3.3.3 A proof by Goursat

**Definition 3.48.** The triangle spanned by  $z_1, z_2, z_3 \in \mathbb{C}$  is the point set

$$\Delta := \left\{ \sum_{j=1}^3 t_j z_j : t_j \geq 0, \sum_{j=1}^3 t_j = 1 \right\}. \quad (3.27)$$

The triangular path  $\langle z_1, z_2, z_3 \rangle$  is the closed curve

$$\langle z_1, z_2, z_3 \rangle = \alpha := \alpha_1 \oplus \alpha_2 \oplus \alpha_3, \quad (3.28)$$

where the edges constituting the triangle boundary are

$$\alpha_1(t) = z_1 + (t-0)(z_2 - z_1), \quad t \in [0, 1];$$

$$\alpha_2(t) = z_2 + (t-1)(z_3 - z_2), \quad t \in [1, 2];$$

$$\alpha_3(t) = z_3 + (t-2)(z_1 - z_3), \quad t \in [2, 3].$$

**Lemma 3.49.** If  $\{\mathcal{P}_i\}$  is a collection of compact subsets of a metric space such that the intersection of every finite subcollection of  $\{\mathcal{P}_i\}$  is nonempty, then  $\cap_i \mathcal{P}_i$  is nonempty.

*Proof.* Denote by  $\mathcal{Q}_i$  the complement of  $\mathcal{P}_i$ . Suppose there exists  $\mathcal{P}_j$  such that no points of  $\mathcal{P}_j$  belongs to every  $\mathcal{P}_i$ . The the sets  $\{\mathcal{Q}_i\}$  is an open cover of  $\mathcal{P}_j$ . By Definition D.145, there exist finitely many indices  $k_1, \dots, k_n$  such that  $\mathcal{P}_j \subset \cup_{\ell=k_1}^{k_n} \mathcal{Q}_\ell$ . But this implies  $\mathcal{P}_j \cap \mathcal{P}_{k_1} \cap \mathcal{P}_{k_2} \cap \dots \cap \mathcal{P}_\ell = \emptyset$ , contradicting the starting point.  $\square$

**Theorem 3.50** (Cantor intersection theorem). Suppose  $(\mathcal{P}_k)$  is a sequence of non-empty, compact subsets of  $\mathcal{P}$  satisfying  $\mathcal{P}_0 \supseteq \mathcal{P}_1 \supseteq \dots \supseteq \mathcal{P}_k \supseteq \dots$ . Then  $\cap_k \mathcal{P}_k \neq \emptyset$ .

*Proof.* This follows directly from Lemma 3.49.  $\square$

**Exercise 3.51.** Prove that if  $\{E_n\}$  is a sequence of closed nonempty and bounded sets in a complete metric space  $\mathcal{X}$ , if  $E_{n+1} \subset E_n$ , and if  $\lim_{n \rightarrow \infty} \text{diam } E_n = 0$ , cf. Definition D.22. Then  $\cap_{n=1}^{\infty} E_n$  consists of exactly one point.

**Exercise 3.52.** Use the conclusion in Exercise 3.51 to prove the Baire category theorem D.80.

**Theorem 3.53** (Cauchy integral theorem on triangular paths). If  $f : \mathcal{D} \rightarrow \mathbb{C}$  is analytic on a domain  $\mathcal{D}$  and the triangle  $\Delta(z_1, z_2, z_3)$  is a subset of  $\mathcal{D}$ , then we have

$$\oint_{\langle z_1, z_2, z_3 \rangle} f(\zeta) d\zeta = 0. \quad (3.29)$$

*Proof.* As shown in the plot below, we first construct a sequence of triangular paths

$$\alpha^{(n)} = \left\langle z_1^{(n)}, z_2^{(n)}, z_3^{(n)} \right\rangle$$

by inductively defining

$$\alpha^{(0)} := \alpha = \langle z_1, z_2, z_3 \rangle,$$

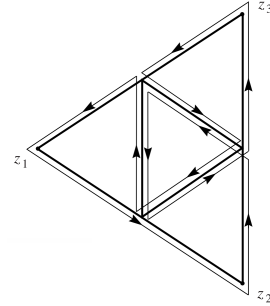
$$\alpha_1^{(n)} := \left\langle \frac{z_1^{(n)} + z_2^{(n)}}{2}, z_2^{(n)}, \frac{z_2^{(n)} + z_3^{(n)}}{2} \right\rangle,$$

$$\alpha_2^{(n)} := \left\langle \frac{z_2^{(n)} + z_3^{(n)}}{2}, z_3^{(n)}, \frac{z_3^{(n)} + z_1^{(n)}}{2} \right\rangle,$$

$$\alpha_3^{(n)} := \left\langle \frac{z_3^{(n)} + z_1^{(n)}}{2}, z_1^{(n)}, \frac{z_1^{(n)} + z_2^{(n)}}{2} \right\rangle,$$

$$\alpha_4^{(n)} := \left\langle \frac{z_1^{(n)} + z_2^{(n)}}{2}, \frac{z_2^{(n)} + z_3^{(n)}}{2}, \frac{z_1^{(n)} + z_3^{(n)}}{2} \right\rangle.$$

and choosing  $\alpha^{(n+1)} = \alpha_1^{(n)}$  or  $\alpha_2^{(n)}$  or  $\alpha_3^{(n)}$  or  $\alpha_4^{(n)}$ .



It follows that

$$\oint_{\alpha^{(n)}} = \oint_{\alpha_1^{(n)}} + \oint_{\alpha_2^{(n)}} + \oint_{\alpha_3^{(n)}} + \oint_{\alpha_4^{(n)}}$$

and we can choose  $\alpha^{(n+1)}$  such that

$$\left| \oint_{\alpha^{(n)}} f \right| \leq 4 \left| \oint_{\alpha^{(n+1)}} f \right|,$$

which implies

$$\left| \oint_{\alpha} f \right| \leq 4^n \left| \oint_{\alpha^{(n)}} f \right|.$$

The closed triangles  $\Delta^{(n)}$  are nested and Exercise 3.51 states that there exists a point  $z_0 \in \cap_n \Delta^{(n)} \subset \mathcal{D}$ . By Lemma 2.18, the condition of  $f$  being analytic at  $z_0$  yields  $f(z) = f(z_0) + f'(z_0)(z - z_0) + r(z)$  where

$$\lim_{z \rightarrow z_0} \frac{r(z)}{|z - z_0|} = 0. \quad (3.30)$$

Then we have  $\oint_{\alpha^{(n)}} f(\zeta) d\zeta = \oint_{\alpha^{(n)}} r(\zeta) d\zeta$  from Theorem 3.16 and Exercise 2.22. It follows that

$$\left| \oint_{\alpha} f(\zeta) d\zeta \right| \leq 4^n \left| \oint_{\alpha^{(n)}} r(\zeta) d\zeta \right|. \quad (3.31)$$

We show  $\lim_{n \rightarrow \infty} 4^n \left| \oint_{\alpha^{(n)}} r(\zeta) d\zeta \right| = 0$  to complete the proof. By (3.30), we have

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } |z - z_0| < \delta \Rightarrow |r(z)| \leq \epsilon |z - z_0|, \quad (3.32)$$

where the condition  $|z - z_0| < \delta$  holds for any sufficiently large  $n$ . In other words, we can choose  $n$  large enough such that  $\Delta^{(n)} \subset \mathcal{U}_\delta(z_0)$ . Denote by  $\ell(\alpha)$  the arc length of  $\alpha$  as

in Definition C.382. Then by construction of the sequence of triangles, we have

$$|z - z_0| \leq \ell(\alpha^{(n)}) = \frac{1}{2^n} \ell(\alpha),$$

which, together with (3.31), (3.32), and Lemma 3.7, yields

$$\left| \oint_{\alpha} f(\zeta) d\zeta \right| \leq 4^n \epsilon \frac{1}{2^n} \ell(\alpha) \ell(\alpha^{(n)}) = \epsilon \ell^2(\alpha).$$

To sum up,  $\left| \oint_{\alpha} f(\zeta) d\zeta \right|$  can be made arbitrarily small, hence (3.29) holds.  $\square$

**Corollary 3.54.** If  $f : \mathcal{D} \rightarrow \mathbb{C}$  is analytic in a simply connected domain  $\mathcal{D}$ , then for any piecewise linear simple closed curve  $\beta \subset \mathcal{D}$  we have

$$\oint_{\beta} f(\zeta) d\zeta = 0. \quad (3.33)$$

*Proof.* This follows from Theorem 3.53 and the two ears theorem, which states that every simple polygon with more than three vertices has at least two ears.  $\square$

**Theorem 3.55** (Heine 1872). If  $K \subset \mathbb{C}$  is compact and  $f : K \rightarrow \mathbb{C}$  is continuous, then  $f$  is uniformly continuous.

*Proof.* This is a specialization of Theorem D.150.  $\square$

**Lemma 3.56.** If  $f : \mathcal{D} \rightarrow \mathbb{C}$  is continuous in a domain  $\mathcal{D}$ , then for any piecewise smooth simple curve  $\alpha \subset \mathcal{D}$  and any  $\epsilon > 0$ , there exists a piecewise linear curve  $\sigma \subset \mathcal{D}$  such that

$$\left| \int_{\alpha} f(\zeta) d\zeta - \int_{\sigma} f(\zeta) d\zeta \right| < \epsilon. \quad (3.34)$$

In other words, the integral of  $f$  on a piecewise smooth curve can be approximated arbitrarily well by an integral of  $f$  on line segments.

*Proof.* Since  $\mathcal{D}$  is open, there exists a compact subset  $G \subset \mathcal{D}$  such that  $\alpha \cap \partial G = \emptyset$  and  $\alpha \subset G$ . Define

$$\rho = \inf\{|z_1 - z_2| : z_1 \in \alpha, z_2 \in \partial G\}.$$

Clearly  $\rho > 0$ . We have  $\forall z \in \alpha, \forall r < \rho, \mathcal{U}_r(z) \subset G$  and

$$\forall z_1, z_2 \in \alpha, |z_1 - z_2| < \rho \Rightarrow \{z_1 t + z_2(1-t) : t \in [0, 1]\} \subset G.$$

By Theorem 3.55, the continuity of  $f$  and the compactness of  $G$  yield the uniform continuity of  $f$  on  $G$ , i.e.,

$$\forall \epsilon > 0, \exists \delta_1 > 0 \text{ s.t. } \forall z_1, z_2 \in G,$$

$$|z_1 - z_2| < \delta_1 \Rightarrow |f(z_1) - f(z_2)| < \frac{\epsilon}{2\ell}, \quad (3.35)$$

where  $\ell$  is the arc length of  $\alpha$ .

Choose  $n + 1$  points  $z_1, z_2, \dots, z_{n+1} \in \alpha$  such that  $\max_{j=1}^n |z_j - z_{j+1}| < \min(\rho, \delta_1)$ : if  $\alpha$  is closed then we set  $z_{n+1} = z_1$ ; if  $\alpha$  is not closed we set  $z_1$  and  $z_{n+1}$  as the starting and ending points, respectively. We also require that  $\alpha^{-1}(z_1), \alpha^{-1}(z_2), \dots, \alpha^{-1}(z_{n+1})$  increases monotonically. Then the piecewise linear closed curve  $\sigma$  formed by connecting  $z_j$ 's consecutively by linear segments is contained

in  $\mathcal{D}$ . Its edges  $\sigma_j$ 's correspond to the arcs  $\alpha_j$ 's. For each  $j$ , Lemma 3.14 yields

$$\int_{\sigma_j} f(z_j) d\zeta = f(z_j)(z_{j+1} - z_j) = \int_{\alpha_j} f(z_j) d\zeta. \quad (3.36)$$

Consequently, we have

$$\begin{aligned} & \left| \int_{\alpha} f(\zeta) d\zeta - \int_{\sigma} f(\zeta) d\zeta \right| \\ & \leq \left| \int_{\alpha_j} [f(\zeta) - f(z_j)] d\zeta \right| + \left| \int_{\sigma_j} [f(\zeta) - f(z_j)] d\zeta \right| \\ & \leq \ell(\alpha_j) \sup_{\zeta \in \alpha_j} |f(\zeta) - f(z_j)| + \ell(\sigma_j) \sup_{\zeta \in \sigma_j} |f(\zeta) - f(z_j)| \\ & < \frac{\epsilon}{2\ell} [\ell(\alpha_j) + \ell(\sigma_j)] \leq \frac{\epsilon}{\ell} \ell(\alpha_j), \end{aligned}$$

where the first step follows from (3.36) and the triangle inequality, the second step from Lemma 3.7 and Definition C.382, and the third step from (3.35). Then we have

$$\begin{aligned} \left| \int_{\alpha} f(\zeta) d\zeta - \int_{\sigma} f(\zeta) d\zeta \right| &= \left| \sum_{j=1}^n \int_{\alpha_j} f(\zeta) d\zeta - \int_{\sigma_j} f(\zeta) d\zeta \right| \\ &\leq \sum_{j=1}^n \left| \int_{\alpha_j} f(\zeta) d\zeta - \int_{\sigma_j} f(\zeta) d\zeta \right| \\ &= \epsilon, \end{aligned}$$

which completes the proof.  $\square$

**Theorem 3.57** (Cauchy integral theorem for simply connected domains). If  $f : \mathcal{D} \rightarrow \mathbb{C}$  is analytic in a simply connected domain  $\mathcal{D}$ , then for any piecewise smooth closed curve  $\alpha \subset \mathcal{D}$  we have

$$\oint_{\alpha} f(\zeta) d\zeta = 0. \quad (3.37)$$

*Proof.* This follows directly from Corollary 3.54 and Lemma 3.56.  $\square$

**Exercise 3.58.** For an analytic function  $f : \mathbb{C} \rightarrow \mathbb{C}$  satisfying  $\forall z \in \mathbb{C}, f(z+1) = f(z)$ , prove that the value of  $\int_{[w, w+1]} f(\zeta) d\zeta$  is independent of the choice of  $w \in \mathbb{C}$ .

**Exercise 3.59.** For bounded simply connected domain  $\mathcal{D}$ ,

- (a) prove  $\oint_{\partial \mathcal{D}} \bar{z} dz = 2iS(\mathcal{D})$ , where  $S(\mathcal{D})$  is the area of  $\mathcal{D}$ ,
- (b) show that  $f$  being analytic on  $\bar{\mathcal{D}}$  implies

$$\max_{z \in \partial \mathcal{D}} |\bar{z} - f(z)| \geq 2 \frac{S(\mathcal{D})}{\ell(\partial \mathcal{D})},$$

where  $\partial \mathcal{D}$  is the boundary of  $\mathcal{D}$  and  $\ell(\partial \mathcal{D})$  its arc length.

**Definition 3.60.** A *star-shaped domain* is an open set  $\mathcal{D} \subset \mathbb{C}$  that satisfies

$$\exists z_* \in \mathcal{D} \text{ s.t. } \forall z \in \mathcal{D}, \{z_* + t(z - z_*) : t \in [0, 1]\} \subset \mathcal{D},$$

where  $z_*$  is called a *star center*.

**Example 3.61.** Each convex domain, in particular the open disk, is star-shaped. The ring sector

$$\{z_0 + \zeta \rho e^{i\varphi} : \rho \in (r, R), \varphi \in (0, \beta); \zeta, z_0 \in \mathbb{C}, |\zeta| = 1\}$$

is star-shaped if  $\beta < \pi$  and  $\cos \frac{\beta}{2} > \frac{r}{R}$ .

**Theorem 3.62** (Cauchy integral theorem for star-shaped domains). Each analytic function  $f$  defined on a star-shaped domain  $\mathcal{D}$  has a primitive in  $\mathcal{D}$ .

*Proof.* This follows from Theorems 3.57 and 3.40 since a star-shaped domain is a simply connected domain.  $\square$

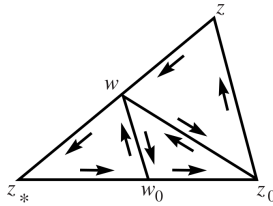
**Exercise 3.63** (Alternative proof of Corollary 3.15). Prove Corollary 3.15 by Theorems 3.62 and 3.40.

**Corollary 3.64.** Let  $f : \mathcal{D} \rightarrow \mathbb{C}$  be continuous in a star-shaped domain  $\mathcal{D}$  with center  $z_*$ . If  $f$  is analytic at every point  $z \neq z_*$ , then  $f$  has a primitive in  $\mathcal{D}$ .

*Proof.* By Theorem 3.40, it suffices to show

$$\oint_{\langle z, z_0, z_* \rangle} f(\zeta) d\zeta = 0,$$

where  $z \neq z_*$  and  $z_0 \neq z_*$  are two arbitrary points in  $\mathcal{D}$  and the triangle is entirely contained within  $\mathcal{D}$ .



As shown in the above plot, Theorem 3.53 yields

$$\oint_{\langle z_*, z_0, z \rangle} = \oint_{\langle z_*, w_0, w \rangle} + \oint_{\langle w_0, z_0, w \rangle} + \oint_{\langle z_0, z, w \rangle} = \oint_{\langle z_*, w_0, w \rangle}.$$

The proof is completed by passing to the limit  $w \rightarrow z_*$  and  $w_0 \rightarrow z_*$ .  $\square$

### 3.3.4 Multiply connected domains

**Theorem 3.65** (Jordan curve theorem). The complement of a Jordan curve  $\gamma$  consists of two components: one is bounded and the other is unbounded; both of them are open and each has  $\gamma$  as its boundary.

**Definition 3.66.** A Jordan curve is said to be *positively oriented* or *counterclockwise oriented* iff an observer traversing in the direction of its parametrization always has the bounded complement on her left; it is *negatively oriented* or *clockwise oriented* iff the observer always has the unbounded complement on her left.

**Definition 3.67.** The *interior* of an oriented Jordan curve  $\gamma$ , denoted by  $\text{int}(\gamma)$ , is the complement of  $\gamma$  that always lies to the left of an observer who traverses the curve in the increasing direction of its parameter; the other complement of  $\gamma$  is called the *exterior* of  $\gamma$  and denoted by  $\text{ext}(\gamma)$ ,

**Lemma 3.68** (Seifert decomposition). Any oriented closed curve can be decomposed into the union of a finite number of oriented, pairwise non-crossing Jordan curves.

*Proof.* The local neighborhood of each self-intersection contains an even number of arc pieces. Rearrange them so that the resulting Jordan curves have no proper intersections.  $\square$

**Definition 3.69.** A *disk with  $n$  holes* ( $n \in \mathbb{N}$ ) is a connected open set  $\mathcal{D} \subset \mathbb{C}$  that can be expressed as

$$\mathcal{D} := \bigcap_{j=0}^n \text{int}(\gamma_j), \quad (3.38)$$

where  $\text{int}(\gamma)$  is the interior of a Jordan curve  $\gamma$  in Definition 3.67,  $\gamma_0$  is a counter-clockwise oriented Jordan curve, and  $\gamma_1, \dots, \gamma_n$  are clockwise oriented Jordan curves satisfying

$$\forall j = 1, \dots, n, \text{ext}(\gamma_j) \subset \text{int}(\gamma_0), \quad (3.39a)$$

$$\forall j, k = 1, \dots, n, j \neq k \Rightarrow \text{ext}(\gamma_j) \cap \text{ext}(\gamma_k) = \emptyset. \quad (3.39b)$$

The boundary and closure of a disk with holes are denoted respectively by

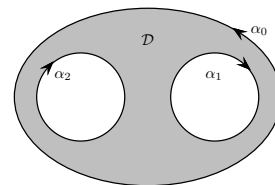
$$\partial\mathcal{D} := \bigcup_{j=0}^n \gamma_j, \quad \overline{\mathcal{D}} := \mathcal{D} \cup \partial\mathcal{D}.$$

**Example 3.70.** It can be shown that any regular open semianalytic set with bounded boundary is a disk with holes. For example, an open disk is a disk with no holes.

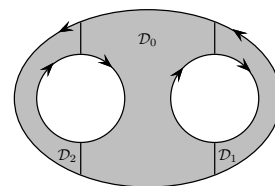
**Theorem 3.71** (Cauchy integral theorem for a disk with holes). Let  $\mathcal{D} := \bigcap_{j=0}^n \text{int}(\gamma_j)$  be a disk with holes as in Definition 3.69. If a complex function  $f$  is analytic on  $\mathcal{D}$  and continuous on  $\overline{\mathcal{D}} := \mathcal{D} \cup \partial\mathcal{D}$ , then we have

$$\int_{\partial\mathcal{D}} f(z) dz = 0.$$

*Proof.* The domain  $\mathcal{D}$  is exemplified for  $n = 2$  as follows.



It follows from a simple induction that  $\bigcup_{j=0}^n \gamma_j$  can be reorganized into positively-oriented Jordan curves with disjoint interiors  $\mathcal{D}_0, \mathcal{D}_1, \dots, \mathcal{D}_m$  as follows.



The proof is then completed by  $\bigcup_{j=0}^n \gamma_j = \bigcup_{j=0}^m \partial\mathcal{D}_j$  and the Cauchy integral theorem 3.57.  $\square$

### 3.4 The Cauchy integral formulas

**Lemma 3.72.** Define a counter-clockwise oriented circle as

$$\alpha(t) = z_0 + re^{it}, \quad z_0 \in \mathbb{C}, t \in [0, 2\pi], r > 0. \quad (3.40)$$

Then we have

$$\forall z \in \mathbb{C}, |z - z_0| < r \Rightarrow \oint_{\alpha} \frac{d\zeta}{\zeta - z} = 2\pi i. \quad (3.41)$$

*Proof.* Referring to Definition 3.69, we construct a disk  $\mathcal{D}$  with one hole by setting  $\gamma_0$  as the circle  $\alpha$  and  $\gamma_1$  as a clockwise oriented circle of which the center is  $z$  and the radius is  $\rho \leq r - |z_0 - z|$ . Theorem 3.71 yields

$$0 = \int_{\partial \mathcal{D}} = \oint_{\gamma} + \oint_{\gamma_1} \Rightarrow \oint_{\gamma} = \oint_{\gamma_1^-}.$$

The rest of the proof follows from Corollary 3.17.  $\square$

**Theorem 3.73** (Cauchy integral formula). Let  $\mathcal{D} \subseteq \mathbb{C}$  be open and  $f : \mathcal{D} \rightarrow \mathbb{C}$  be an analytic function. Suppose that the closed disk  $\overline{\mathcal{U}}_r(z_0)$  is completely contained in  $\mathcal{D}$ . Then

$$\forall z \in \mathcal{U}_r(z_0), \quad f(z) = \frac{1}{2\pi i} \oint_{\alpha} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad (3.42)$$

where  $\alpha$  is the circle defined in (3.40).

*Proof.* By the compactness of  $\overline{\mathcal{U}}_r(z_0)$ , there exists an  $R > r$  such that

$$\overline{\mathcal{U}}_r(z_0) \subset \mathcal{U}_R(z_0) \subset \mathcal{D}.$$

We can thus assume that  $\mathcal{D}$  is an open disk. The function

$$g(w) := \begin{cases} \frac{f(w) - f(z)}{w - z} & \text{if } w \neq z, \\ f'(z) & \text{if } w = z \end{cases}$$

is analytic in  $\mathcal{D} \setminus \{z\}$  and is continuous in  $\mathcal{D}$ . Then Corollary 3.64 and Theorem 3.40 yield

$$\oint \frac{f(\zeta) - f(z)}{\zeta - z} d\zeta = 0 \Rightarrow f(z) \oint \frac{1}{\zeta - z} d\zeta = \oint \frac{f(\zeta)}{\zeta - z} d\zeta.$$

The proof is completed by Lemma 3.72.  $\square$

**Corollary 3.74** (Cauchy integral formula for non-simply connected domains). Let  $\gamma_0$  be a counter-clockwise oriented Jordan curve and  $\gamma_1, \dots, \gamma_n$  be clockwise oriented Jordan curves satisfying (3.39). If a complex function  $f$  is analytic on the disk with holes  $\mathcal{D} := \cap_{j=0}^n \text{int}(\gamma_j)$  and continuous on  $\overline{\mathcal{D}}$ , then we have

$$\oint_{\gamma_0} f(z) dz = \sum_{j=1}^n \oint_{\gamma_j^-} f(z) dz, \quad (3.43)$$

where each  $\gamma_j^-$  is  $\gamma_j$  with its orientation reversed, cf. Definition 3.26.

*Proof.* This follows from Theorem 3.71 by moving the summation term in (3.43) to the right-hand side (RHS) and reversing the orientation of  $\gamma_j$ 's for each  $j = 1, 2, \dots, n$ .  $\square$

**Example 3.75.** Let  $\alpha$  denote the circle defined in (3.40) with  $z_0 = 0$  and  $r = 2$ . We compute

$$\oint_{\alpha} \frac{\zeta}{(9 - \zeta^2)(\zeta + i)} d\zeta = \oint_{\alpha} \frac{\frac{\zeta}{9 - \zeta^2}}{\zeta - (-i)} d\zeta = 2\pi i \left. \frac{\zeta}{9 - \zeta^2} \right|_{\zeta = -i} = \frac{\pi}{5},$$

where the second step follows from Theorem 3.73.

**Corollary 3.76** (Mean value equation). For the special case of  $z = z_0$ , Theorem 3.73 reduces to

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt. \quad (3.44)$$

*Proof.* This follows directly from Theorem 3.73 by substituting  $z = z_0$  and  $\zeta = z_0 + re^{it}$ .  $\square$

**Example 3.77.** To compute  $I = \oint_{|z|=2} \frac{dz}{z^2 + 2}$ , we construct two counter-clockwise oriented circles  $\alpha_1$  and  $\alpha_2$  surrounding the two singularities  $z = \pm\sqrt{2}i$ . For sufficiently small radii, Corollary 3.74 yields

$$\begin{aligned} \oint_{|z|=2} f(z) dz &= \oint_{\alpha_1} f(z) dz + \oint_{\alpha_2} f(z) dz \\ &= \oint_{\alpha_1} \frac{1}{z - \sqrt{2}i} dz + \oint_{\alpha_2} \frac{1}{z - (-\sqrt{2}i)} dz \\ &= \left. \frac{2\pi i}{z + \sqrt{2}i} \right|_{z=\sqrt{2}i} + \left. \frac{2\pi i}{z - \sqrt{2}i} \right|_{z=-\sqrt{2}i} = 0. \end{aligned}$$

**Exercise 3.78.** Compute  $I = \oint_{|z|=1} \frac{3z-1}{z(z-2)} dz$ .

**Exercise 3.79.** For some  $r > 0$ ,  $f$  is an analytic function on  $\{z \in \mathbb{C} : |z| > r\}$ . Show that  $\lim_{z \rightarrow \infty} zf(z) = A$  implies

$$\forall R > r, \quad \oint_{|z|=R} f(z) dz = 2\pi i A.$$

**Exercise 3.80.** For  $R > 0$ , write  $\mathcal{U}_R := \{z \in \mathbb{C} : |z| < R\}$ .

(a) Let  $f : \mathcal{U}_{R_0} \rightarrow \mathbb{C}$  be an analytic function, where  $R_0 > 0$ . Prove that whenever  $R \in (0, R_0)$  and  $z \in \mathcal{U}_R$ , then

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(Re^{it}) \frac{R^2 - |z|^2}{|Re^{it} - z|^2} dt.$$

(b) Show that

$$\text{Re} \left( \frac{Re^{it} + z}{Re^{it} - z} \right) = \frac{R^2 - |z|^2}{|Re^{it} - z|^2}.$$

(c) If  $u$  is harmonic in the unit disk  $\mathbb{D} = \mathcal{U}_1$  and continuous on  $\overline{\mathbb{D}}$ , then show that for  $z \in \mathbb{D}$ ,

$$u(z) = \int_0^{2\pi} P(z, e^{it}) u(e^{it}) dt,$$

where  $P(z, e^{it})$  is the *Poisson kernel* defined by

$$P(z, e^{it}) = \frac{1}{2\pi} \frac{1 - |z|^2}{|e^{it} - z|^2} = \frac{1}{2\pi} \text{Re} \left( \frac{e^{it} + z}{e^{it} - z} \right).$$

*Hint:* For (a), note that if  $w = \frac{R^2}{\bar{z}}$ , then the integral of  $\frac{f(\zeta)}{\zeta - w}$  over the circle  $|\zeta| = R$  is zero. Use this, together with the usual Cauchy integral formula, to deduce the desired result. For (c), apply Exercise 3.39, then use (a) and (b).

**Theorem 3.81** (Leibniz integral rule). Let  $f : [a, b] \times [c, d] \rightarrow \mathbb{R}$  be a continuous function. Suppose that the partial derivative  $\frac{\partial f(t, x)}{\partial x}$  exists and is continuous. Also suppose that the functions  $\alpha, \beta : [c, d] \rightarrow [a, b]$  are both continuously differentiable. Then

$$\begin{aligned} & \frac{d}{dx} \left( \int_{\alpha(x)}^{\beta(x)} f(t, x) dt \right) \\ &= \int_{\alpha(x)}^{\beta(x)} \frac{\partial f(t, x)}{\partial x} dt + f(\beta(x), x) \frac{d\beta(x)}{dx} - f(\alpha(x), x) \frac{d\alpha(x)}{dx}. \end{aligned} \quad (3.45)$$

*Proof.* This follows from the chain rule (Theorem B.88).  $\square$

**Lemma 3.82.** Let  $\mathcal{D} \subseteq \mathbb{C}$  be open and  $f : [a, b] \times \mathcal{D} \rightarrow \mathbb{C}$  be a continuous function that is also analytic in  $\mathcal{D}$  for any fixed  $t \in [a, b]$ . Also assume that the partial derivative  $\frac{\partial f}{\partial z} : [a, b] \times \mathcal{D} \rightarrow \mathbb{C}$  is continuous. Then the function

$$g(z) := \int_a^b f(t, z) dt$$

is analytic in  $\mathcal{D}$  and

$$g'(z) = \int_a^b \frac{\partial f(t, z)}{\partial z} dt. \quad (3.46)$$

*Proof.* For any fixed  $t$ , write  $f(t, z) = p(t, x, y) + iq(t, x, y)$  and  $g(z) = u(x, y) + iv(x, y)$ . Then we have  $u = \int p dt$  and  $v = \int q dt$ . Then the Cauchy–Riemann theorem 2.50 applied to  $f$  yields

$$\begin{aligned} \frac{\partial u}{\partial x} &= \int \frac{\partial p}{\partial x} = \int \frac{\partial q}{\partial y} = \frac{\partial v}{\partial y}, \\ \frac{\partial u}{\partial y} &= \int \frac{\partial p}{\partial y} = - \int \frac{\partial q}{\partial x} = - \frac{\partial v}{\partial x}. \end{aligned}$$

Since  $\frac{\partial f}{\partial z} : [a, b] \times \mathcal{D} \rightarrow \mathbb{C}$  is continuous,  $g(z)$  is totally differentiable and thus analytic by the above equations and the Cauchy–Riemann theorem. The formula (3.46) follows from Theorem 3.81.  $\square$

**Theorem 3.83** (Generalized Cauchy integral formula). Every analytic function is arbitrarily often complex differentiable. Each derivative is again analytic. Furthermore,

$$f^{(n)}(z) = \frac{n!}{2\pi i} \oint_{\alpha} \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta, \quad (3.47)$$

where the assumptions and notation are the same as those in Theorem 3.73.

*Proof.* Define

$$G(t, z) := \frac{f(\zeta(t))\zeta'(t)}{\zeta(t) - z}.$$

From Theorem 3.73 we have

$$f(z) = \frac{1}{2\pi i} \oint_{\alpha} G(t, z) dt.$$

The rest of the proof follows from Lemma 3.82 and an induction on  $n$ .  $\square$

**Corollary 3.84** (Cauchy inequality). If  $f : \mathcal{D} \rightarrow \mathbb{C}$  is analytic and  $\mathcal{U}_r(z_0) \subset \mathcal{D}$  for some  $r > 0$ , then we have

$$\forall n \in \mathbb{N}^+, \quad |f^{(n)}(z_0)| \leq \frac{n!M(r)}{r^n}, \quad (3.48)$$

where  $M(r) = \max_{|z-z_0|=r} |f(z)|$ .

*Proof.* For the circle defined in (3.40), Theorem 3.83 yields

$$\begin{aligned} |f^{(n)}(z_0)| &= \left| \frac{n!}{2\pi i} \oint_{\alpha} \frac{f(\zeta)}{(\zeta - a)^{n+1}} d\zeta \right| \\ &\leq \frac{n!}{2\pi} \frac{M(r)}{r^{n+1}} 2\pi r = \frac{n!M(r)}{r^n}. \end{aligned} \quad \square$$

**Example 3.85.** For a Jordan curve  $\alpha$  around  $\mathbf{i}$ , we have

$$\begin{aligned} \oint_{\alpha} \frac{\cos z}{(z - \mathbf{i})^3} dz &= \frac{2\pi \mathbf{i}}{2!} (\cos z)''|_{z=\mathbf{i}} = -\pi \mathbf{i} \cos \mathbf{i} \\ &= -\pi \frac{e^{-1} + e}{2} \mathbf{i}. \end{aligned}$$

**Exercise 3.86.** Compute  $I = \oint_{|z|=2\pi} \frac{z^3}{(z-\pi)^3} dz$ .

**Exercise 3.87.** Suppose  $f : \mathbb{D} \rightarrow \mathbb{C}$  is analytic. Show that the diameter  $d = \sup_{z, w \in \mathbb{D}} |f(z) - f(w)|$  of the image of  $f$  satisfies

$$2|f'(0)| \leq d.$$

*Hint:* Show that  $2f'(0) = \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{f(\zeta) - f(-\zeta)}{\zeta^2} d\zeta$  for any  $r \in (0, 1)$ .

**Exercise 3.88.** Let  $f$  be a bounded analytic function on the unit disk  $\mathbb{D}$ . Prove that

$$\sup_{z \in \mathbb{D}} (1 - |z|) |f'(z)| < +\infty.$$

*Hint:* For  $z \in \mathbb{D}$ , use the Cauchy inequality at the circle centered at  $z$  with radius  $1 - |z|$ .

**Exercise 3.89.** Let  $f$  be an analytic function on the strip  $\mathcal{D} := \{z \in \mathbb{C} : |\operatorname{Im} z| < 1\}$ . For a fixed  $\eta \in \mathbb{R}$ , there exists  $A > 0$  such that

$$\forall z \in \mathcal{D}, \quad |f(z)| \leq A(1 + |z|)^{\eta}.$$

Show that for each  $n \in \mathbb{N}$ , there exists  $A_n > 0$  such that

$$\forall x \in \mathbb{R}, \quad |f^{(n)}(x)| \leq A_n(1 + |x|)^{\eta}.$$

*Hint:* Use the Cauchy inequality.

**Exercise 3.90.** Let  $f$  be an analytic function on the closed unit disk  $\overline{\mathbb{D}}$ .

(a) Prove that

$$\frac{1}{2\pi i} \oint_{|\zeta|=1} \frac{\overline{f(\zeta)}}{\zeta - z} d\zeta = \begin{cases} \overline{f(0)} & \text{if } |z| < 1, \\ \overline{f(0) - f(\frac{1}{\bar{z}})} & \text{if } |z| > 1. \end{cases}$$

(b) If  $|z| < 1$ , compute

$$\frac{1}{2\pi i} \oint_{|\zeta|=1} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

(c) If  $P$  is a polynomial, show that

$$\oint_{|\zeta|=1} \overline{P(\zeta)} d\zeta = 2\pi i \overline{P'(0)}.$$

*Hint:* You can verify that

- $\int_{\alpha} \overline{f(\zeta)} d\zeta = \overline{\int_{\alpha} f(\zeta) d\bar{\zeta}}$ , where  $\alpha$  is any piecewise smooth curve.
- For  $|\zeta| = 1$ , it holds that  $\bar{\zeta} = \frac{1}{\zeta}$ ,  $d\bar{\zeta} = -\frac{d\zeta}{\zeta^2}$  and  $d|\zeta| = \frac{d\zeta}{i\zeta}$ .

### 3.5 Morera's and Liouville's theorems

**Theorem 3.91** (Morera 1886). Let  $\mathcal{D} \subseteq \mathbb{C}$  be open and  $f : \mathcal{D} \rightarrow \mathbb{C}$  be a continuous function. If

$$\oint_{\langle z_1, z_2, z_3 \rangle} f(\zeta) d\zeta = 0 \quad (3.49)$$

for every triangular path  $\langle z_1, z_2, z_3 \rangle$  whose triangle is entirely contained in  $\mathcal{D}$ , then  $f$  is analytic.

*Proof.* For any given point  $z_0 \in \mathcal{D}$ , there is an open neighborhood  $\mathcal{U}_\epsilon(z_0) \subset \Delta(z_1, z_2, z_3) \subset \mathcal{D}$ . It suffices to show that  $f$  is analytic in  $\mathcal{U}_\epsilon(z_0)$ . By (3.49), Lemma 3.56, and the fact that every polygon has a triangulation, the integral of  $f$  along any closed curve in  $\mathcal{U}_\epsilon(z_0)$  vanishes. Theorem 3.40 then implies that  $f$  has a primitive  $F$  and the proof is completed by applying Theorem 3.83 to  $F$ .  $\square$

**Definition 3.92.** An analytic function  $f : \mathbb{C} \rightarrow \mathbb{C}$  is said to be *entire*.

**Theorem 3.93** (Liouville 1847). Every bounded entire function is a constant.

*Proof.* We show  $f'(z) = 0$  for all  $z \in \mathbb{C}$ . Since  $f$  is bounded, there exists  $C > 0$  such that  $|f(z)| \leq C$  for all  $z \in \mathbb{C}$ . Then Corollary 3.84 implies that

$$\forall z \in \mathbb{C}, \forall r > 0, \quad |f'(z)| \leq \frac{C}{r}.$$

Passing to the limit  $r \rightarrow +\infty$  completes the proof.  $\square$

**Example 3.94.**  $f(z) = \cos(z)$  is not bounded since

$$\lim_{x \rightarrow +\infty} \cos(ix) = \lim_{x \rightarrow +\infty} \frac{e^x + e^{-x}}{2} = +\infty.$$

**Exercise 3.95** (The generalized Liouville). Let  $f$  be an entire function. For a given integer  $m \geq 1$ , there exists  $C > 0$  such that

$$\forall z \in \mathbb{C}, \quad |f(z)| \leq C(1 + |z|)^m.$$

Show that  $f$  is a polynomial of degree at most  $m$ .

*Hint:* Use the Cauchy inequality to show  $f^{(m+1)} \equiv 0$ .

**Exercise 3.96.** We say  $f : \mathbb{C} \rightarrow \mathbb{C}$  is *doubly periodic function* if there exist  $\tau \in \mathbb{C} \setminus \mathbb{R}$  such that

$$\forall z \in \mathbb{C}, \quad f(z+1) = f(z+\tau) = f(z).$$

Show that every entire doubly periodic function is a constant.

**Exercise 3.97.** Let  $f$  be a nonconstant entire function. Show that  $f(\mathbb{C})$  is dense in  $\mathbb{C}$ , that is, for any  $\zeta \in \mathbb{C}$ , there exists a sequence  $\{z_n\}$  such that  $f(z_n) \rightarrow \zeta$ .

*Hint:* Suppose there exist  $\zeta \in \mathbb{C}$  and  $r > 0$  such that  $\mathcal{U}_r(\zeta) \cap f(\mathbb{C}) = \emptyset$ . Then apply Liouville's theorem to  $\frac{1}{f(z)-\zeta}$ .

**Exercise 3.98** (Alternative proof of the fundamental theorem of algebra). Prove Theorem 3.47 by applying Liouville's theorem.